Near Similarity and Counterfactuals

Abstract

David Lewis (1971; 1973) analyzes counterfactuals in terms of maximal similarity. In this paper, I argue that they should instead be analyzed in terms of near similarity. After giving two arguments for this shift, I develop near-similarity semantics as a positive proposal by providing models and a corresponding proof theory. A striking feature of near-similarity semantics is that it employs a similarity relation that need not be transitive. As a result, it invalidates the inference rule Strengthen Might.

David Lewis (1971; 1973) says that counterfactuals should be analyzed in terms of similarity. In particular, let @ be the actual world.[1] Then:

$A \mathrel{\Box\!\!\rightarrow} B$ iff every most similar A-world to @ is a B-world.
$A \mathrel{\Diamond\!\!\rightarrow} B$ iff some most similar A-world to @ is a B-world.

I agree that counterfactuals should be analyzed in terms of similarity. However, I say that we should analyze them using the *nearly* most similar worlds, not the most similar worlds. So:

$A \mathrel{\Box\!\!\rightarrow} B$ iff every nearly most similar A-world to @ is a B-world.
$A \mathrel{\Diamond\!\!\rightarrow} B$ iff some nearly most similar A-world to @ is a B-world.

We thus have two competing semantic theories. Lewis advances what I will call **maximal-similarity semantics**. I advance **near-similarity semantics**. For short, I will refer to these theories as m-semantics and n-semantics.

Here is the plan for this paper. First, I will present two arguments against m-semantics (§§1–2). I will then develop n-semantics as an alternative by giving models and a proof theory (§§3–4). Finally, I will consider various replies and objections (§§5–7).

[1]For now, we adopt the limit assumption. We will eventually drop this in §3.

1 First Argument

Suppose that Socrates is exactly five feet tall. How tall would he have been, had he been at least six feet? Worlds in which Socrates' height is closer to his actual height are more similar to the actual world. So the most similar worlds in which he is at least six feet are those in which he is exactly six feet. Thus on m-semantics, (1) is true and (2) is false:

Had Socrates been at least six feet tall, he would have been exactly six feet tall. (1)

Had Socrates been at least six feet tall, he might have been more than six feet tall. (2)

But this is the wrong result. In fact, (1) is false and (2) is true.[2]

In contrast, n-semantics gives the right result. It may be that the most similar worlds in which Socrates is at least six feet are those in which he is exactly six feet. Still, there are worlds in which he is just a bit taller that are *nearly* as similar. So the nearly most similar worlds include worlds in which Socrates is more than six feet. So (1) is false and (2) is true.

2 Second Argument

Planck lengths are incredibly small. You would quite literally need a hundred quintillion of them just to span the diameter of a proton. Now suppose that Socrates is exactly five feet. For each positive integer n, let s_n be a sentence saying that Socrates is at least n Planck lengths tall. Let i be the number of Planck lengths corresponding to six feet. Let k be the number of Planck lengths corresponding to nine feet. We then claim that:

Boundedness: $s_i \mathrel{\Box\!\!\rightarrow} \neg s_k$

This says that had Socrates been at least six feet, he would have been less than nine feet. Furthermore, we claim that for all j such that $i \leq j \leq k$:

[2] Hájek (2025) independently raises a similar objection. His response is to abandon similarity semantics altogether. I say that this is unnecessary, since we can instead move from m-semantics to n-semantics. So while Hájek and I agree that there is a problem, we disagree about the solution.

Smoothness:	$s_j \diamondright s_{j+1}$
Monotonicity:	$s_{j+1} \boxright s_j$

Both of these principles are schemas.[3] The first says that for each j between six feet and nine feet (inclusive), had Socrates been at least j Planck lengths, he might have been at least $j+1$ Planck lengths. So for example, had Socrates been at least six feet, he might have been at least six feet and one Planck length. The second says that for each j between six feet and nine feet (inclusive), had Socrates been at least $j+1$ Planck lengths, he would thereby have been at least j Planck lengths. Thus, had Socrates been at least six feet and one Planck length, he would thereby have been at least six feet.

With these premises in place, we can derive a flat contradiction using propositional logic and four axioms:

SP:	$(A \wedge B \boxright C) \rightarrow (B \wedge A \boxright C)$
LT:	$[(A \boxright B) \wedge (A \wedge B \boxright C)] \rightarrow [A \boxright C]$
SM:	$[(A \boxright C) \wedge (A \diamondright B)] \rightarrow [A \wedge B \boxright C]$
DL:	$(A \boxright B) \equiv \neg(A \diamondright \neg B)$

These axioms are Swap, Limited Transitivity, Strengthen Might, and Duality. All four are accepted by Lewis. All four are validated by m-semantics. We now reason as follows:

1.	$s_i \boxright \neg s_k$	Boundedness
2.	$s_i \diamondright s_{i+1}$	Smoothness
3.	$s_i \wedge s_{i+1} \boxright \neg s_k$	1, 2, SM
4.	$s_{i+1} \boxright s_i$	Monotonicity
5.	$s_{i+1} \wedge s_i \boxright \neg s_k$	3, SP
6.	$s_{i+1} \boxright \neg s_k$	4, 5, LT

After $k-i-1$ iterations of this reasoning, we eventually reach $s_{k-1} \boxright \neg s_k$. But then:

1.	$s_{k-1} \boxright \neg s_k$	
2.	$s_{k-1} \diamondright s_k$	Smoothness
3.	$\neg(s_{k-1} \boxright \neg s_k)$	2, DL
4.	$\perp$	1, 3, LNC

So we have reasoned to absurdity. Something has to go.

[3] We are using schemas because our official language is a purely propositional language, and thus has no quantifiers.

This argument shows that in m-semantics, the premises are inconsistent. Thus, Lewis is committed to rejecting at least one of them. However, the premises would seem to be not only consistent but true. At the end of the next section, we will show that they are consistent in n-semantics. So n-semantics gives the right result.

3 Models

We thus have two arguments for n-semantics. Later on, I will consider replies and objections. For that, it will be helpful to have more tools on the table. Thus, I will set aside replies and objections for now and turn to giving models for both m-semantics and n-semantics. In the next section, I will give proof theories.

Our official language $\mathcal{L}$ will have atomic sentences $p, q, r, \dots$ and the operators $\neg$, $\wedge$ and $\Box\!\!\rightarrow$. We then define:

$$
\begin{aligned}
A \vee B &:= \neg(\neg A \wedge \neg B) & \top &:= p \vee \neg p \\
A \rightarrow B &:= \neg(A \wedge \neg B) & \bot &:= \neg\top \\
A \equiv B &:= (A \rightarrow B) \wedge (B \rightarrow A) & \Box A &:= \neg A \mathrel{\Box\!\!\rightarrow} \bot \\
A \mathrel{\Diamond\!\!\rightarrow} B &:= \neg(A \mathrel{\Box\!\!\rightarrow} \neg B) & \Diamond A &:= \neg\Box\neg A
\end{aligned}
$$

I think of $\Box$ as expressing a special notion of necessity tied to counterfactuals. Call this notion **counterfactual necessity**.

Definition 3.1: A **frame** $\langle W, \preceq\rangle$ is a pair with $W \neq \varnothing$ and $\preceq \subseteq W^3$. For convenience, we write $w \preceq_x v$ when $\langle w, v, x\rangle \in \preceq$.

A frame consists of a set of worlds W and a three-place accessibility relation $\preceq$ on those worlds. We will generally think of $\preceq$ as a similarity relation on worlds. Formally speaking though, it could be any ternary relation on W.

Definition 3.2: The **asymmetric part** $\prec$ and the **symmetric part** $\approx$ of an accessibility relation $\preceq$ are defined with:

$$
w \prec_x v := (w \preceq_x v) \wedge (v \not\preceq_x w) \qquad w \approx_x v := (w \preceq_x v) \wedge (v \preceq_x w)
$$

Definition 3.3: A **model** $\langle W, \preceq, [\![\cdot]\!]\rangle$ is a frame together with an interpretation $[\![\cdot]\!] : \mathsf{Sent}(\mathcal{L}) \to \mathcal{P}(W)$ satisfying:

$$\begin{aligned}
[\![\neg A]\!] &= W \setminus [\![A]\!] \\
[\![A \wedge B]\!] &= [\![A]\!] \cap [\![B]\!] \\
[\![A \mathrel{\Box\!\!\rightarrow} B]\!] &= \{x \in W \mid [[\![A]\!] = \varnothing] \vee [\exists w \in [\![A]\!]\, \forall v \in [\![A \wedge \neg B]\!]\, (\, w \prec_x v\,)]\}
\end{aligned}$$

A model consists of a frame together with an interpretation $[\![\cdot]\!]$ that maps each sentence in our language to a set of worlds, which you might think of as a proposition. I will say more about how to read the clause for $\Box\!\!\rightarrow$ shortly.[4]

Our models give us a general framework for theorizing about counterfactuals in terms of similarity. We then build m-models and n-models by adding further constraints on accessibility. These constraints (strong centering, moderate centering, total preorder, semiorder) will be defined shortly. For now, I just want to have the basic definitions of m-models and n-models in place.

Definition 3.4: A **maximal-similarity frame** (m-frame) is a frame in which the accessibility relation is a strongly centered total preorder.

Definition 3.5: A **near-similarity frame** (n-frame) is a frame in which the accessibility relation is a moderately centered semiorder.

Definition 3.6: A **maximal-similarity model** (m-model) is a model based on an m-frame.

Definition 3.7: A **near-similarity model** (n-model) is a model based on an n-frame.

When using m-models, we can think of the accessibility relation as an **as-similar** relation. Thus:

$w \preceq_x v$	w is at least as similar as v to x
$w \prec_x v$	w is more similar than v to x

[4]The frames and models described here are similar to those described by Burgess (1981), who also uses a ternary accessibility relation. However, there are two key differences. First, where Burgess requires the accessibility relation to be a preorder, we do not. Second, Burgess uses a more complex semantic clause for $\Box\!\!\rightarrow$, which is needed for classes of frames in which totality fails. Since m-frames and n-frames are total, we can use a simpler clause. For more on this, see Observation 1.3 in (Author 2026b).

$w \approx_x v$ w and v are equally similar to x

In contrast, when using n-models, we can think of the accessibility relation as a **nearly-as-similar** relation. So:

$w \preceq_x v$ w is at least nearly as similar as v to x
$w \prec_x v$ w is much more similar than v to x
$w \approx_x v$ w and v are nearly equally similar to x

Intuitively then, m-models and n-models differ only with respect to which kind of similarity relations they use for accessibility. Because these similarity relations have different formal properties, m-models and n-models generate different counterfactual logics.

How should we read the clause for $\Box\!\!\rightarrow$ in definition 3.3? This depends on which models we are using. If we are using m-models:

> $A \Box\!\!\rightarrow B$ is true at a world x iff either there is no A-world or there is some A-world that is more similar to x than every A-world that is not a B-world. (3)

If we are using n-models:

> $A \Box\!\!\rightarrow B$ is true at a world x iff either there is no A-world or there is some A-world that is much more similar to x than every A-world that is not a B-world. (4)

Thus, the two kinds of models use the same basic semantic clauses. The main difference is how we think about the asymmetric part $\prec$ of the accessibility relation $\preceq$. When using m-models, we think of $\prec$ as the relation of being more similar. When using n-models, we think of $\prec$ as the relation of being *much* more similar.

The clauses for the counterfactual operators that I gave in the introduction assume that we have the limit assumption in place:

$$\text{Limit Assumption:} \quad [[\![A]\!] \neq \varnothing] \rightarrow [\exists w \in [\![A]\!]\, \forall v \in [\![A]\!]\, (v \nprec_x w)]$$

For m-models, the limit assumption says that whenever there is at least one A-world, there is at least one most similar A-world.[5] For n-models, it says that whenever there is at least one A-world, there is at least one nearly most similar A-world.

Does the limit assumption hold? I am ambivalent. Consider an alleged counterexample from Lewis (1973: pp. 20–21): Suppose that you draw a line on a piece of paper that is one inch long. Worlds at which the line is 1/2 inch longer are more similar than worlds in which it is 1 inch longer; worlds at which it is 1/4 inch longer are more similar than worlds in which it is 1/2 inch longer; and so on. Thus, there are no most similar worlds at which the line is longer than one inch. So the limit assumption fails.

Perhaps this counterexample works if we are using m-semantics. It does not work, however, if we are using n-semantics. In order to have a counterexample to the limit assumption when using n-semantics, we would need an infinite descending sequence of worlds, each of which is *much* more similar than the last. I claim, however, that Lewis has not given us any such sequence. For example, worlds in which the line is 1/2048 inches longer than one inch may be more similar to the actual world than worlds in which it is 1/1024 inches longer. Still, they are not *much* more similar. So Lewis has not given us a counterexample.

Is there a counterexample that works for n-semantics? Suppose that the actual world is one in which there are $\aleph_0$ many stars. That is, suppose that the actual world is one in which the number of stars is countably infinite. We now claim that worlds with 10^2 stars are much more similar to the actual world than worlds with 10^1 stars, that worlds with 10^3 stars are much more similar than worlds with 10^2 stars, and so on and so forth. So the limit assumption fails.

Does this counterexample work? I am not entirely sure. It depends on how we measure similarity. Let w_{n+1} be a world with exactly 10^{n+1} stars and let w_n be an otherwise similar world with exactly 10^n stars. One could say that w_{n+1} is much more similar to @ than w_n, on the grounds that w_{n+1} matches the actual world on vastly more star-facts. But one could instead say that they are

[5]When using m-models, the set of most similar A-worlds to x is $\{w \in [\![A]\!] \mid \forall v \in [\![A]\!](v \not\prec_x w)\}$. When using n-models, the set of nearly most similar A-worlds is defined in exactly the same way.

equally similar to @, on the grounds that $\aleph_0 - 10^n = \aleph_0 - 10^{n+1} = \aleph_0$.[6] For my own part, I have no intuitions about which of these similarity measures is the right one.

Fortunately, there is no need to take a stand on the limit assumption for present purposes.[7] All I want to do is make three points: First, counterexamples to the limit assumption in m-semantics need not be counterexamples to the limit assumption in n-semantics. Second, the semantic clause for $\Box\!\!\rightarrow$ from definition 3.3 entails the clauses for both $\Box\!\!\rightarrow$ and $\Diamond\!\!\rightarrow$ from the introduction. This is true for both m-models and n-models. Finally, the clause from definition 3.3 does not require the limit assumption, unlike the clauses from the introduction. Thus, the clause from definition 3.3 is more general.

I will now say more about the conditions used to define m-models and n-models.[8] A **total preorder** is a relation that is total and transitive:

Total:	$(w \preceq_x v) \vee (v \preceq_x w)$
Transitive:	$[(w \preceq_x v) \wedge (v \preceq_x u)] \rightarrow [w \preceq_x u]$

As-similar relations are both total and transitive.[9,10] Thus, they are total preorders. Nearly-as-similar relations are also total.[11] However, they need not be transitive, and so need not be total preorders.[12]

[6]Here $\kappa - \lambda$ is a notion of cardinal subtraction, defined with $\kappa - \lambda := \min\{\mu : \kappa \leq \lambda + \mu\}$. Intuitively, $\kappa - \lambda$ is the least cardinal μ such that adding μ to λ suffices to reach at least κ. This operation is well-defined for all $\lambda \leq \kappa$.

[7]The formal reason for this is that the class of all m-frames generates the same counterfactual logic as the class of all m-frames that satisfy the limit assumption. Likewise for the class of all n-frames. See (Author 2026b) for further discussion.

[8]For displayed properties, like the ones here, variables range over the domain of worlds and are bound by implicit wide-scope universal quantifiers.

[9]Take any worlds w, v, and x. Either w is at least as similar as v to x or v is at least as similar as w to x. So as-similar relations are total. They are also transitive. After all, take any worlds w, v, u, and x. If w is at least as similar as v to x and v is at least as similar as u to x, then w is at least as similar as u to x.

[10]Totality is controversial for as-similar relations, and will be controversial for nearly-as-similar relations for similar reasons. See Pollock (1976) for a view on which it fails for as-similar relations. I will not defend totality here, since totality is something that both Lewis and I accept. I agree with Lewis's reply to Pollock, which can be found in (Lewis 1981)226.

[11]Take any worlds w, v, and x. Either w is at least nearly as similar as v to x or v is at least nearly as similar as w to x.

[12]Properties like totality and transitivity are most often thought of as properties of binary relations. Here, we are thinking of them as properties of ternary relations. There is, however, a close connection between the two kinds of properties. Say that the binary relation $\preceq^*$ is a *slice* of the ternary relation $\preceq$ when for some x and all w and all v, $w \preceq^* v$ iff $w \preceq_x v$. A ternary

To illustrate, suppose that Socrates is five feet at the actual world @. We then construct a sequence of worlds $w_0, w_1, w_2, \ldots, w_m$. Socrates is six feet at w_0, six feet and one Planck length at w_1, six feet and two Planck lengths at w_2, and so on. Finally, he is m Planck lengths taller than six feet at w_m, where m Planck lengths equals three feet. Thus, Socrates is exactly nine feet at w_m. Each world w_{n+1} in the sequence is at least nearly as similar as w_n to @. If the relevant nearly-as-similar relation were transitive, then w_m would be at least nearly as similar as w_0 to @. But this is false. In fact, w_0 is much more similar than w_m to @.

Nearly-as-similar relations need not be transitive. Still, they are semitransitive.[13] This property consists of three conditions:

$$\text{Semitransitive:}\quad \begin{array}{l} [(w \prec_x v) \wedge (v \prec_x u) \wedge (u \preceq_x t)] \rightarrow [w \prec_x t] \\ [(w \prec_x v) \wedge (v \preceq_x u) \wedge (u \prec_x t)] \rightarrow [w \prec_x t] \\ [(w \preceq_x v) \wedge (v \prec_x u) \wedge (u \prec_x t)] \rightarrow [w \prec_x t] \end{array}$$

Semitransitivity may look complex, but the basic idea is simple. Take the first condition. This says that if w is much more similar than v to x, and v is much more similar than u to x, and u is at least nearly as similar as t to x, then w is much more similar than t to x. The other two conditions are analogous. The only difference is that the pattern of strict and non-strict relations in the antecedent has been permuted.

We said earlier that a total preorder is a relation that is total and transitive. In contrast, a relation is a **semiorder** when it is total and semitransitive.[14,15] Thus, where as-similar relations are total preorders, nearly-as-similar relations are semiorders.

This brings us to centering conditions. Centering conditions place constraints on how similar worlds are to themselves. Such conditions include:

relation is then reflexive when every slice is reflexive, transitive when every slice is transitive, and so on and so forth.

[13] Semitransitivity has no well-established name in the literature. Candeal et al. (2002) call this property generalized pseudotransitivity.

[14] Semiorders are usually credited to Luce (1956). However, they were in fact first introduced almost forty years earlier by Wiener (1914), a mathematician who studied under Bertrand Russell.

[15] There are different kinds of semiorders. What I am here calling semiorders are weak semiorders. Besides these, there are also strict semiorders. Moreover, the two are closely related. A relation $\preceq$ is a weak semiorder iff the asymmetric part $\prec$ is a strict semiorder. See (Author 2026b) for further discussion.

$$
\begin{array}{ll}
\text{Weak Centering:} & x \preceq_x w \\
\text{Moderate Centering:} & (v \prec_x w) \to (x \prec_x w) \\
\text{Strong Centering:} & (x \neq w) \to (x \prec_x w)
\end{array}
$$

Lewis thinks that as-similar relations are strongly centered, and this is plausible.[16] In contrast, nearly-as-similar relations need not be strongly centered.

To show this, let x be a world in which Socrates is five feet and let w be a world that is exactly the same, except that Socrates is one Planck length taller. While x may be more similar than w to x, it is not *much* more similar. So strong centering fails. But while strong centering fails, moderate centering still holds. If there is a world v that is much more similar than w to x, then x is also much more similar than w to x.

To close this section, I want to return to the argument from §2. There, we showed that the premises are jointly inconsistent on m-semantics. Now that we have models, we can show that they are jointly consistent on n-semantics.

Proposition 3.8: *Boundedness, Smoothness, and Monotonicity are jointly satisfiable in the class of all n-models.*

Proof. Let Γ be the set of sentences containing Boundedness, along with every instance of Smoothness and Monotonicity. We then construct a model $\mathfrak{M}$ as follows. Let w^* be the number of Planck lengths corresponding to five feet. We then set:

$$
\begin{aligned}
W &= \mathbb{N} \\
\preceq &= \{\langle w, v, x\rangle \in W^3 \mid |w - x| - |v - x| \leq 1\} \\
[\![s_n]\!] &= \{w \in W \mid w \geq n\}
\end{aligned}
$$

It is straightforward to verify that $\mathfrak{M}$ is an n-model and that $\mathfrak{M}, w^* \vDash \Gamma$. ■

[16] After all, take any $w \neq x$. Since $w \neq x$, there is some difference between w and x. There is no difference between x and x. So x is more similar than w to x.

4 Logic

Lewis accepts a system of counterfactual logic that I am going to call **M**.[17] To build this system, we add two rules and nine axioms to classical propositional logic:

$$
\begin{array}{ll}
\text{LE:} & \text{If } \vdash A \equiv B, \text{ then } \vdash (A \mathrel{\Box\!\!\rightarrow} C) \equiv (B \mathrel{\Box\!\!\rightarrow} C) \\
\text{RW:} & \text{If } \vdash A \rightarrow B, \text{ then } \vdash (C \mathrel{\Box\!\!\rightarrow} A) \rightarrow (C \mathrel{\Box\!\!\rightarrow} B) \\
\text{ID:} & A \mathrel{\Box\!\!\rightarrow} A \\
\text{DI:} & ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C)) \rightarrow (A \vee B \mathrel{\Box\!\!\rightarrow} C) \\
\text{CO:} & ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Box\!\!\rightarrow} B)) \rightarrow (A \mathrel{\Box\!\!\rightarrow} (B \wedge C)) \\
\text{SW:} & ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Box\!\!\rightarrow} B)) \rightarrow (A \wedge B \mathrel{\Box\!\!\rightarrow} C) \\
\text{SM:} & ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow (A \wedge B \mathrel{\Box\!\!\rightarrow} C) \\
\text{SC:} & (A \wedge B) \rightarrow (A \mathrel{\Box\!\!\rightarrow} B) \\
\text{T:} & \Box A \rightarrow A \\
\text{4:} & \Box A \rightarrow \Box\Box A \\
\text{5:} & \Diamond A \rightarrow \Box\Diamond A
\end{array}
$$

The two rules are Left Equivalence and Right Weakening. The first six axioms are Identity, Disjunction, Conjunction, Strengthen Would, Strengthen Might, and Strong Centering. The last three are familiar modal axioms. These together ensure that the logic of counterfactual necessity is **S5**.[18]

The result is a system that is sound and strongly complete with respect to the class of all m-frames.[19] Thus, you might think of **M** as the counterfactual logic of maximal similarity.

While I agree with Lewis about most of the above principles, we disagree about SM and SC. He accepts both. I reject both. Thus, in the rest of this section, I will discuss each of these axioms in turn. I will then say what I think

[17]**M** is equivalent to what Lewis (1973) calls **VCU**, which is the result of adding the modal axioms 4 and 5 to **VC**. Lewis says that **VC** is his official system, not because he in fact denies S4 and S5, but because he does not want to "exaggerate" his disagreements with Stalnaker, whose preferred system **VCS** does not include 4 and 5 (Lewis 1973)99. However, Lewis himself seems to be ambivalent.

For the purposes of this paper, I am attributing **VCU** to Lewis, rather than **VC**, primarily because this lets us use simpler models. For more general models, which allow us to model weaker systems like **VC**, see (Author 2026b).

[18]Note that there is no need to add RN as a basic rule, since RN follows from ID and LE, given the definition of $\Box$.

[19]See Theorem 8.5 in (Author 2026b), where the system **M** is referred to as **GTPs** and the class of all m-frames is the class of all strongly centered global total preorder models.

should be put in their place. Finally, I will describe my own preferred system, which I call **N**.

4.1 Against SM

When can the antecedent of a would counterfactual be strengthened? One answer is that it can always be strengthened, under any conditions whatsoever. This is what Full Strengthening says:

$$\text{FS:}\quad (A \mathrel{\Box\!\!\rightarrow} C) \rightarrow (A \wedge B \mathrel{\Box\!\!\rightarrow} C)$$

Lewis says that FS fails, and I agree. After all, suppose that had Socrates been at least six feet, he would have been no more than seven feet. Does it follow that had he been at least six feet *and* at least seven feet, he would have been exactly seven feet? This is equivalent to saying that had he been at least seven feet, he would have been exactly seven feet. But this does not follow. Had Socrates been at least seven feet, he might have been a little more than seven feet.

I say that SM fails for similar reasons. Suppose that had Socrates been at least six feet, he would have been no more than seven feet. Suppose also that had he been at least six feet, he might have been at least seven feet. Does it follow that had he been at least six feet *and* at least seven feet, he would have been exactly seven feet? This is equivalent to saying that had Socrates been at least seven feet, he would have been exactly seven feet. But this does not follow. Had he been at least seven feet, he might have been a little more than seven feet.[20]

SM is valid in m-semantics because as-similar relations are always transitive. In contrast, SM is not valid in n-semantics precisely because nearly-as-similar relations need not be transitive.

To show that SM is invalid in n-semantics, suppose that we have an n-model in which there are exactly four possible worlds, as illustrated in figure 1. Bars between worlds indicate that those worlds are nearly equally similar to x. Arrows indicate that the target world is much more similar than the source world. We now assign values to the atomic sentences p, q, and r as illustrated. You can then confirm that $p \mathrel{\Box\!\!\rightarrow} r$ and $p \mathrel{\Diamond\!\!\rightarrow} q$ are true at x, but $p \wedge q \mathrel{\Box\!\!\rightarrow} r$ is false

[20] Others have also rejected Strengthen Might, though for different reasons. See for example Boylan and Schultheis (2021).

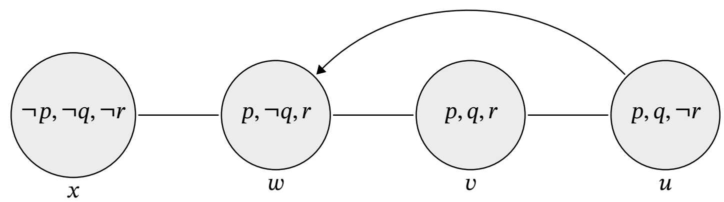

Figure 1: A Countermodel to SM

Alt text: A diagram of four possible worlds arranged in a row from left to right and labeled x, w, v, and u. World x assigns $\neg p, \neg q, \neg r$; world w assigns $p, \neg q, r$; world v assigns p, q, r; and world u assigns $p, q, \neg r$. Plain edges connect adjacent worlds (x to w, w to v, and v to u), indicating that each adjacent pair is nearly equally similar to x. A curved directed arrow from u to w indicates that w is much more similar than u to x.

at x. So we have a countermodel to SM. Moreover, we have a countermodel precisely because transitivity fails (since $u \preceq_x v$ and $v \preceq_x w$ but $u \npreceq_x w$).

4.2 Against SC

When is a might counterfactual true? Lewis and I agree that $\diamondto$ can be defined in terms of $\boxto$. Thus, we agree that:

$$A \diamondto B := \neg(A \boxto \neg B) \tag{5}$$

In that case, given classical logic, SC is equivalent to Modus Ponens for Might:

$$\text{MPM:} \quad (A \wedge (A \diamondto B)) \to B$$

But this is obviously invalid. Suppose that Alice is considering a run for president. You point out that if she were to run, she might win. So she runs. Does it follow that she will win? Clearly not.[21]

Lewis could respond by giving up (5). This would let him keep SC while denying MPM. In that case, though, Lewis no longer has a theory of might counterfactuals, and so will need to give us a different one. This is obviously a cost.

Moreover, even if Lewis were to deny (5), there are arguably direct counterexamples to SC. To borrow a case from McDermott (2007), suppose that you

[21] See (Author 2026a) for further discussion of MPM.

bet that a certain coin will land heads twice. The coin is then flipped twice and lands heads twice. So you win! After the flipping, you meet a friend, who mistakenly thinks that the coin landed tails twice. She remarks that:

> If the coin had landed heads at least once, you would have won. (6)

This would seem to be false. The coin could have easily landed heads at least once because it landed heads *only* once. In that case, you would not have won. So SC fails.

SC is valid given m-semantics because as-similar relations are strongly centered. After all, no world could be as similar to the actual world as the actual world is to itself. In contrast, SC fails in near-similarity semantics because nearly-as-similar relations need not be strongly centered. After all, it may be that no world is as similar to the actual world as the actual world is to itself. Still, some worlds are *nearly* as similar to the actual world as the actual world is to itself.[22]

4.3 The Logic of Near Similarity

If SM and SC are invalid, what should we put in their place? I say that we should replace them with:

$$\begin{array}{ll}
\text{SEM:} & [A \mathrel{\Box\!\!\rightarrow} C] \rightarrow [(A \wedge B \mathrel{\Box\!\!\rightarrow} C) \vee (A \wedge \neg B \mathrel{\Box\!\!\rightarrow} C)] \\
\text{RSM:} & [(A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B) \wedge (A \wedge \neg C \mathrel{\Box\!\!\rightarrow} \neg B)] \rightarrow [A \wedge B \mathrel{\Box\!\!\rightarrow} C] \\
\text{MC:} & [(A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \wedge B)] \rightarrow [A \wedge B \mathrel{\Box\!\!\rightarrow} C]
\end{array}$$

These axioms are Strengthen Excluded Middle, Restricted Strengthen Might, and Moderate Centering. All three govern the strengthening of antecedents.

Why accept these axioms? The short answer is that they would seem to be valid. Suppose that had Alice gone to the party, she would have had a good time. SEM then tells us that at least one of two things is true: Either (a) had Alice gone to the party with Ben, she would have had a good time or (b) had Alice gone to the party without Ben, she would have had a good time.

[22] The model illustrated in figure 1 is also a countermodel to SC. In this model, x and w are both among the nearly most similar $\neg q$ worlds to x. The sentences $\neg q$ and $\neg p$ are both true at x. However, $\neg q \mathrel{\Box\!\!\rightarrow} \neg p$ is false at x, since $\neg q$ is true at w and $\neg p$ is false at w.

Next, suppose that had Margaux taken the exam, she would have studied. Moreover, had she taken the exam, she might have received a perfect score. But of course, had she taken the exam and *not* studied, she would *not* have received a perfect score. RSM then tells us that had Margaux taken the exam and received a perfect score, she would have studied.

Finally, suppose that if the rocket were launched, it would safely reach the moon. Moreover, suppose that the rocket is in fact launched with a lunar lander. In that case, MC tells us that if the rocket were launched with a lunar lander, it would safely reach the moon.

This brings us to my own preferred system, which I call **N**. To build this system, we start with the list of principles we used to build **M**. We then simply delete SM and SC while adding SEM, RSM, and MC. Everything else stays the same. Since SEM, RSM, and MC are also theorems of **M**, the resulting system **N** is strictly weaker.[23] This system is sound and strongly complete with respect to the class of all n-frames.[24] Thus, you might think of **N** as the counterfactual logic of near similarity.

5 Returning to the First Argument

We have now developed n-semantics as a positive view by building models (§3) and a proof theory (§4). Along the way, we also gave models and a proof theory for m-semantics. With these additional tools in place, I now want to consider potential replies to the two arguments I gave to start the paper (§§1–2). This section will focus on the argument from §1. The next will focus on the argument from §2.

Could Lewis respond to the first argument by appealing to vagueness? After all, it can be somewhat vague how tall Socrates is at each world, and somewhat vague which similarity relation we are using in any given context.

None of this helps though. Determinately, worlds at which Socrates is closer to his actual height are more similar to the actual world. So deter-

[23]Pollock (1976) advances a semantics on which similarity relations are not required to be total. The result is that SM fails. His semantics, however, also invalidates SEM and RSM, which I think is clearly the wrong result. Kratzer (1981) advances an equivalent semantics with the same features. See Lewis (1981) for further discussion.

[24]See Theorem 8.5 in (Author 2026b), where the system **N** is referred to as **GSm** and the class of all n-frames corresponds to the class of all moderately centered global total semiorder frames.

minately, the most similar worlds in which he is at least six feet are worlds in which he is exactly six feet. Now of course, it may be somewhat vague *which* worlds are among the most similar worlds. Maybe it is somewhat vague whether Socrates is exactly six feet tall at some particular world, for example. But this is neither here nor there. Since it is determinate that the most similar worlds are ones in which he is exactly six feet, it is determinate that (1) is true and (2) is false. So appealing to vagueness is no help.

Alternatively, Lewis could say that in ordinary contexts, we use loose standards. For example, suppose that we are looking at Jones and Smith. They seem to be the same height, so we agree that:

Jones and Smith are the same height. (7)

Still, Jones and Smith are almost certainly not the same height, strictly speaking. One of them is probably at least a few Planck lengths taller than the other. However, we are not interested in whether Jones and Smith are the same height, down to the Planck length. Instead, we are interested in whether they are the same height in a contextually relevant looser sense.

Lewis could say that the same goes for similarity. In ordinary contexts, we count worlds as being equally similar in the loose sense even when they are not equally similar in the strict sense. As a result, we count worlds in which Socrates is a bit over six feet as being just as similar as worlds in which he is exactly six feet. This means that in ordinary contexts, m-semantics predicts that (1) is false and (2) is true, which is the right result.

In response, I am happy to grant that we often speak loosely. Still, we also speak strictly. For example, suppose that you have a friend who claims that Jones and Smith are the same height. You might grant that this is true loosely speaking. Still, you point out that strictly speaking, Jones is six feet and Smith is six feet and a quarter inch. Thus, they may be the same height loosely speaking, but they are not the same height strictly speaking.

Now consider a parallel case involving counterfactuals. You say that had Socrates been at least six feet, he might have been more than six feet. Your friend grants that this is true loosely speaking. Still, she points out that strictly speaking, there is a world in which Socrates is exactly six feet that is more similar than any world at which he is more than six feet. Thus, strictly speaking, it is false that had Socrates been at least six feet, he might have been more than six feet.

I say that your friend is clearly wrong. Not only is it true loosely speaking that had Socrates been at least six feet, he might have been more than six feet. It is true *strictly speaking* that had Socrates been at least six feet, he might have been more than six feet. Loose standards have nothing to do with it.

I take this point to be more or less decisive. However, suppose for the sake of argument that we always speak loosely. For one reason or another, it is simply impossible to speak strictly. Still, I say, loose standards will not solve the problem for Lewis.

We can start by distinguishing between two kinds of loose standards. On the one hand, there are what I will call **blurred standards**. These are standards according to which things are equal in the loose sense when they are nearly equal in the strict sense. So for example, if we are using blurred standards for height, we might say that:

> x is at least as tall as y in the loose sense iff x is at least nearly as tall as y in the strict sense. (8)

If we were using blurred standards for similarity, we would say that:

> w is at least as similar as v to x in the loose sense iff w is at least nearly as similar as v to x in the strict sense. (9)

Besides blurred standards, there are also **rounded standards**. These are standards according to which things are equal in the loose sense when they are equal in the strict sense, after rounding down to an appropriate value. If we were using rounded standards for height, we might say that:

> x is at least as tall as y in the loose sense iff x is at least as tall as y in the strict sense after rounding down to the closest inch. (10)

If we were using rounded standards for similarity, we might say that:

> w is at least as similar as v to x in the loose sense iff the difference in the height of Socrates at w and x is no greater than the difference in the height of Socrates at v and x, when rounding down to the closest inch. (11)

I should say in advance that the rounded standards given in (11) are simplified and for illustrative purposes only. They only work if Socrates exists at every world and has a height at every world and the only thing that matters for similarity is his height. Still, for present purposes, (11) captures the spirit of rounded standards, which is to use rounded heights, rather than precise heights, when forming judgements about the relative similarity of worlds.

The first point I want to make is that while Lewis can perhaps use rounded standards, he cannot use blurred standards. The problem is that blurred as-similar relations need not be strongly centered total preorders. In fact, they may be only moderately centered semiorders. Thus, if Lewis uses blurred standards, he will have to give up his m-models in favor of my n-models, and give up his system **M** in favor of my system **N**. But in that case, Lewis will have simply adopted my view. Game, set, match.

Thus, Lewis cannot use blurred standards. So he will have to use rounded standards. The proposed rounded standards from (11) give us a loose as-similar relation that is a total preorder, so this is a step in the right direction. Still, this rounded as-similar relation is not strongly centered. After all, suppose that Socrates is five feet tall at w and five feet and half an inch at v. Then given our rounded standards, v is at least as similar as w to w. So strong centering fails.

At this point, Lewis could reject strong centering. He could also try to secure strong centering by brute force.[25] For present purposes, we will just assume that this problem has been solved, one way or another. In that case, rounded standards would seem to give Lewis a plausible version of his view on which (1) is false and (2) is true in contexts where we are speaking loosely.

While rounded standards would seem to be promising, they only work if Lewis places strong—and I say arbitrary—constraints on how loose standards work. For my own part, I think that we use both blurred standards and rounded standards. Which standards we use is often vague and context dependent. Still, we can and do use both. This sort of moderate view, though, is not available to Lewis. He has to say that there are no contexts in which we even vaguely use blurred standards. Otherwise, his preferred system **M** fails.[26]

[25]For example, he could define a rounded as-similar relation by stipulating that $x \prec_x w$ whenever $x \neq w$. He would then define the rest of the relation using rounded heights in the normal way.

[26]Suppose there is some context c in which there is some permissible precisification p under which we are using blurred standards for similarity. Then there is some c in which there is

Moreover, even if Lewis can show that we always determinately use rounded standards, he will have to show that we always round in the right direction. After all, if we round *up* to the closest inch, then (1) is true and (2) is false. This is the wrong result, so we must always round down. But *why* do we always round down? Why do we never round up?

Finally, even if we always round down, m-semantics still makes the wrong predictions in a wide range of cases. For example, suppose that the present context is one in which we are rounding down to the closest inch. Thus, worlds in which Socrates is less than six feet are more similar to the actual world than worlds in which he is at least six feet. In that case, m-semantics predicts that (12) is true and (13) is false:

> Had Socrates been around six feet tall, he would have been less than six feet tall. (12)
>
> Had Socrates been around six feet tall, he might have been more than six feet tall. (13)

But this is the wrong result. In fact, (12) is false and (13) is true. Lewis could try to fix the problem by rounding down to different values. Maybe he could say that we round down to the closest centimeter? But in that case, his semantics will make the wrong predictions about different sentences of the same form. This is just moving the bump under the rug.

6 Returning to the Second Argument

The second argument bears a kind of family resemblance to a sorites paradox. Thus, you might wonder whether the argument *just is* a sorites paradox. Because if so, Lewis can say that the problem should be solved in whatever way sorites paradoxes are solved. Thus, there may be a problem to solve, but this is not a problem for his counterfactual semantics.

In response, I say that we can fully precisify our language. Moreover, once we do, the argument only becomes *easier* to make. The same is not true for a sorites paradox. Thus, the second argument is not a sorites paradox.

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some *p* under which some theorem of **M** is false. Genuine logical truths are true in all contexts under all precisifications. So there is some theorem of **M** that is not a genuine logical truth. So **M** is not the right logic for counterfactuals.

To illustrate, suppose that we fully precisify so that for every height h between six feet and seven feet, had Socrates been at least h, he might have been up to three inches taller than h. Now let p be the claim that Socrates is at least six feet, q the claim that he is at least six foot three, and r the claim that he is more than six foot three. Given our precisification:

A1	$\neg(p \Diamond\!\!\rightarrow r)$
A2	$p \Diamond\!\!\rightarrow q$
A3	$q \Box\!\!\rightarrow p$
A4	$q \Diamond\!\!\rightarrow r$

With these four premises in place, we reason as follows:

1.	$\neg(p \Diamond\!\!\rightarrow r)$	A1
2.	$p \Box\!\!\rightarrow \neg r$	1, DL
3.	$p \Diamond\!\!\rightarrow q$	A2
4.	$p \wedge q \Box\!\!\rightarrow \neg r$	2, 3, SM
5.	$q \wedge p \Box\!\!\rightarrow \neg r$	4, SP
6.	$q \Box\!\!\rightarrow p$	A3
7.	$q \Box\!\!\rightarrow \neg r$	5, 6, LT
8.	$q \Diamond\!\!\rightarrow r$	A4
9.	$\neg(q \Diamond\!\!\rightarrow r)$	7, DL
10.	$\bot$	8, 9, LNC

So we have once again derived a contradiction. This time, we used a language with no vagueness. This time, the proof took only ten lines.

In response, Lewis must deny that the precisification I gave is permissible.[27] But on what grounds? Suppose we start with our original vague language. We then precisify so that had Socrates been at least six feet, he might have been up to six foot three. This is obviously permissible. So we have A1, A2, and A3.

[27] Suppose there is a permissible precisification π on which A1–A4 are all true. If the rules of **M** are genuinely valid, then they preserve truth under all permissible precisifications. If the axioms are genuinely valid, then they are true under all permissible precisifications. Thus, if the rules and axioms of **M** are genuinely valid, then π is inconsistent, given the above derivation. Inconsistent precisifications are not permissible. So π is not a permissible precisification. But this is contrary to assumption, so the rules and axioms of **M** are not all genuinely valid.

Note that for the purposes of this argument, it is irrelevant whether there is also some *other* permissible precisification π^* on which the rules and axioms of **M** do not entail a contradiction. All we need is one permissible precisification on which they do.

The question now is about A4. Lewis will have to deny that there is any further permissible precisification on which A4 is true. Otherwise, the gig is up.

But why would there be no such precisification? A4 is not definitely false in our original language. Thus, Lewis will have to say that there is no such precisification because A1, A2, and A3 jointly entail the negation of A4. But this is implausible: It may be that had Socrates been at least six feet, he might have been up to six foot three. Still, it does not follow that had he been at least six foot three, he would have been *exactly* six foot three. In fact, had he been at least six foot three, he might have been a little taller.

7 Strengthen Easy

FS is invalid. Still, instances often sound perfectly fine. For example, suppose that had Margaux stayed home, she would have read a book. It would then seem to follow that had Margaux stayed home and made tea, she would have read a book. The question is, why does this sound like good reasoning, if FS is in fact invalid?

Lewis has a natural answer: He can say that reasoning from $A \mathrel{\Box\!\!\rightarrow} C$ to $A \wedge B \mathrel{\Box\!\!\rightarrow} C$ seems like good reasoning in many contexts because we have $A \mathrel{\Diamond\!\!\rightarrow} B$ as a background assumption. Thus, the reasoning is licensed by SM.

I reject SM. Thus, I need to give a different explanation of why reasoning according to FS often seems good. Moreover, I need to also explain why reasoning according to SM often sounds good. In this final section, I will do both.

Our modal practice includes not only counterfactuals, but **comparative modals**. These are claims like:

$$\text{It could have as easily been that } A \text{ as that } B. \tag{14}$$

So for example, you might think that it could have as easily been that it snowed today as that it rained today, or you might think that it could have as easily been that there was life on Mars as that there was life on Venus.

To regiment such claims, we can add a two-place sentential operator $\trianglelefteq$ to our language. We then read $A \trianglelefteq B$ as saying that it could have as easily been that A as that B.

We now need to give a semantics for the $\trianglelefteq$ operator. To do that, suppose that we start with an n-model. We can then use the nearly-as-similar relation $\preceq$ to define an as-similar relation $\preceq^*$ with:

$$w \preceq^*_x v \coloneqq [w = x] \vee \\ [v \neq x \wedge \forall u \in W\,((v \preceq_x u \rightarrow w \preceq_x u) \wedge (u \preceq_x w \rightarrow u \preceq_x v))] \tag{15}$$

The resulting relation is a strongly centered total preorder. It also interacts with the original nearly-as-similar relation in the ways you would expect.[28] Armed with this new as-similar relation, we can give a semantics for the $\trianglelefteq$ operator:

$$[\![A \trianglelefteq B]\!] \quad = \quad \{x \in W \mid \forall v \in [\![B]\!]\, \exists w \in [\![A]\!]\, (w \preceq^*_x v)\}$$

Thus, it could have as easily been that A as that B at a world x iff for every B-world v, there is an A-world w that is at least as similar as v to x.

This semantics is simple and natural. It also validates an axiom that we will call Strengthen Easy:

$$\text{SE:} \quad [(A \mathrel{\Box\!\!\rightarrow} C) \wedge ((A \wedge B) \trianglelefteq (A \wedge \neg B))] \rightarrow [A \wedge B \mathrel{\Box\!\!\rightarrow} C]$$

Suppose that had Margaux stayed home, she would have read a book. Moreover, suppose that it could have as easily been that she stayed home and made tea as that she stayed home and did not make tea. SE then tells us that had Margaux stayed home and made tea, she would have read a book.

Now that we have SE, we can explain why reasoning according to FS and SM often seems like good reasoning. In the case of FS, we reason from $A \mathrel{\Box\!\!\rightarrow} C$ to $A \wedge B \mathrel{\Box\!\!\rightarrow} C$. In the case of SM, we do the same, using $A \mathrel{\Diamond\!\!\rightarrow} B$ as an additional premise. Both kinds of reasoning can sound perfectly fine. Why is that? I say that this is because we often have $(A \wedge B) \trianglelefteq (A \wedge \neg B)$ as a background assumption. Thus, in such cases, the reasoning in question is in fact licensed by SE.

[28] For example, the proposed definition gives us various mixed principles like:

$$(w \preceq^*_x v) \rightarrow (w \preceq_x v) \qquad (u \preceq^*_x v) \wedge (v \prec_x w) \rightarrow (u \prec_x w)$$
$$(u \preceq^*_x v) \wedge (v \preceq_x w) \rightarrow (u \preceq_x w) \qquad (u \prec_x v) \wedge (v \preceq^*_x w) \rightarrow (u \prec_x w)$$
$$(u \preceq_x v) \wedge (v \preceq^*_x w) \rightarrow (u \preceq_x w)$$

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